Consent

Informed Consent for Online Survey

Study Title: Explanations for Self-Driving Cars.

Purpose of the Study: You are invited to participate in a research study examining the ability of explanations in self-driving cars. You must be at least 18 years of age to participate.

Procedures: If you choose to participate, you will answer some questions after watching short videos pertaining to the actions of a self-driving car. The study is expected to take approximately 15 minutes.

Potential Risks and Benefits: Your participation in the survey will not expose you to harmful, disturbing, or offensive content. There are no specific benefits or risks associated with participating in the survey.

Confidentiality: The data collected in this study is completely anonymous. No personally identifiable information will be collected and the information you choose to provide in this study cannot be connected back to you. Results from this study may be used for any purposes by the survey owner including, to publish or present academic papers. All data collected will be owned by the survey owner. The videos shown during the survey are for testing and experimental purposes only. The videos, and any other information, shown to you during the survey remain the sole and exclusive intellectual property of the survey owner. You are hereby prohibited from using the videos for any purpose other than responding to the survey.

Voluntary Participation: Your participation in this study is voluntary and you may choose to not participate or end your participation at any time without penalty.

Consent: I have read and understand the above consent form. I certify that I am 18 years old or older. By clicking the “Next” button to enter the survey, I indicate my willingness to voluntarily take part in this study.

Preamble

Please do not take this study on a mobile phone, the videos and text will not display correctly.

Please complete the study in one sitting.

The survey should take around 15 minutes.

After you are finished with the survey, you will be redirected to Qualtrics to get paid.

Thank you for your participation!

Prolific ID

Please enter your Prolific ID

Instructions

Instructions

You will see 13 videos of an **autonomous vehicle** (AV) driving in Las Vegas. Each video will contain **real-time**

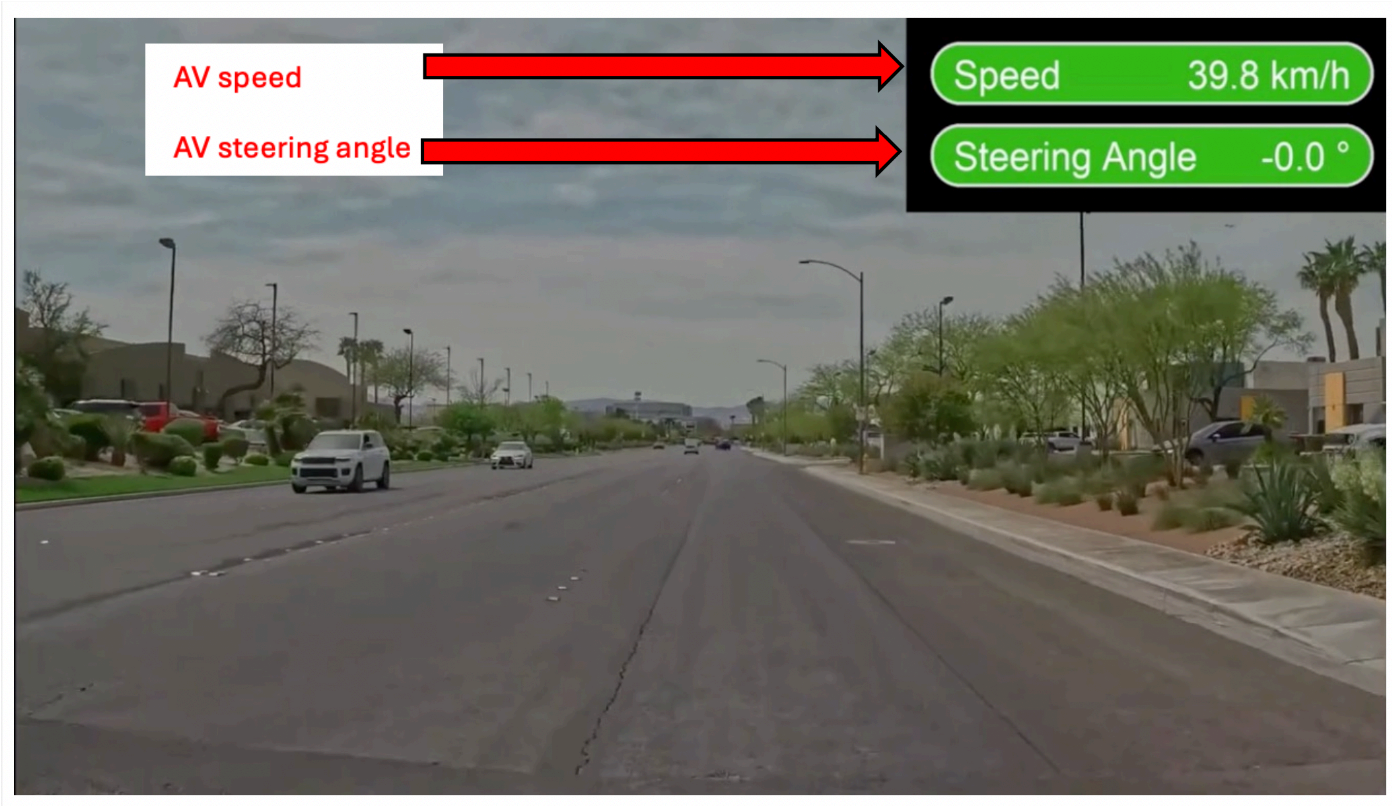
information of the AV behavior. Specifically, you will see its current speed and steering angle:

- Speed is in kilometers per hour (km/h)
- Negative steering angle indicate turning right, positive ones indicate turning left.
- A steering angle of 0.0 indicates the AV is driving straight.
- This information will constantly change throughout the video in real time.

At random points, the video will pause and you will be asked questions.

The two steps are:

Step 1 (Look at the video, speed, and steering angle, and wait for it to pause...)



Step 2 (answer yes/no questions about the video)

Please answer the following questions carefully.

	Yes	No
Was there a straight road in the video?	○	○
Did the AV ever detect a straight road?	○	○
Was there a stopped vehicle in front of the AV at any point?	○	○
Did the AV ever detect a stopped vehicle?	○	○
Was the AV driving straight partly because it detected a straight road?	○	○
If there was a traffic jam ahead, would the AV have stopped?	○	○

Instructions part 2

Final Instructions

Each scenario you see is with a different AV, you should consider them all separately and *DO NOT conflate the ability of one AV to the next AV.* For example, if you think one video demonstrates the AV is very good at detecting a certain concept, you should not assume the AV in the next video is also good at detecting this concept.

Lastly, you will repeatedly be asked if (for example) a **bike** was in the video, and then if the AV ***detected*** the bike. To clarify, just because an object is in front of the AV, it does not mean it ***detected*** it, so take care to differentiate between the two questions.

You will only be asked yes/no questions, if you are not sure, try your best.

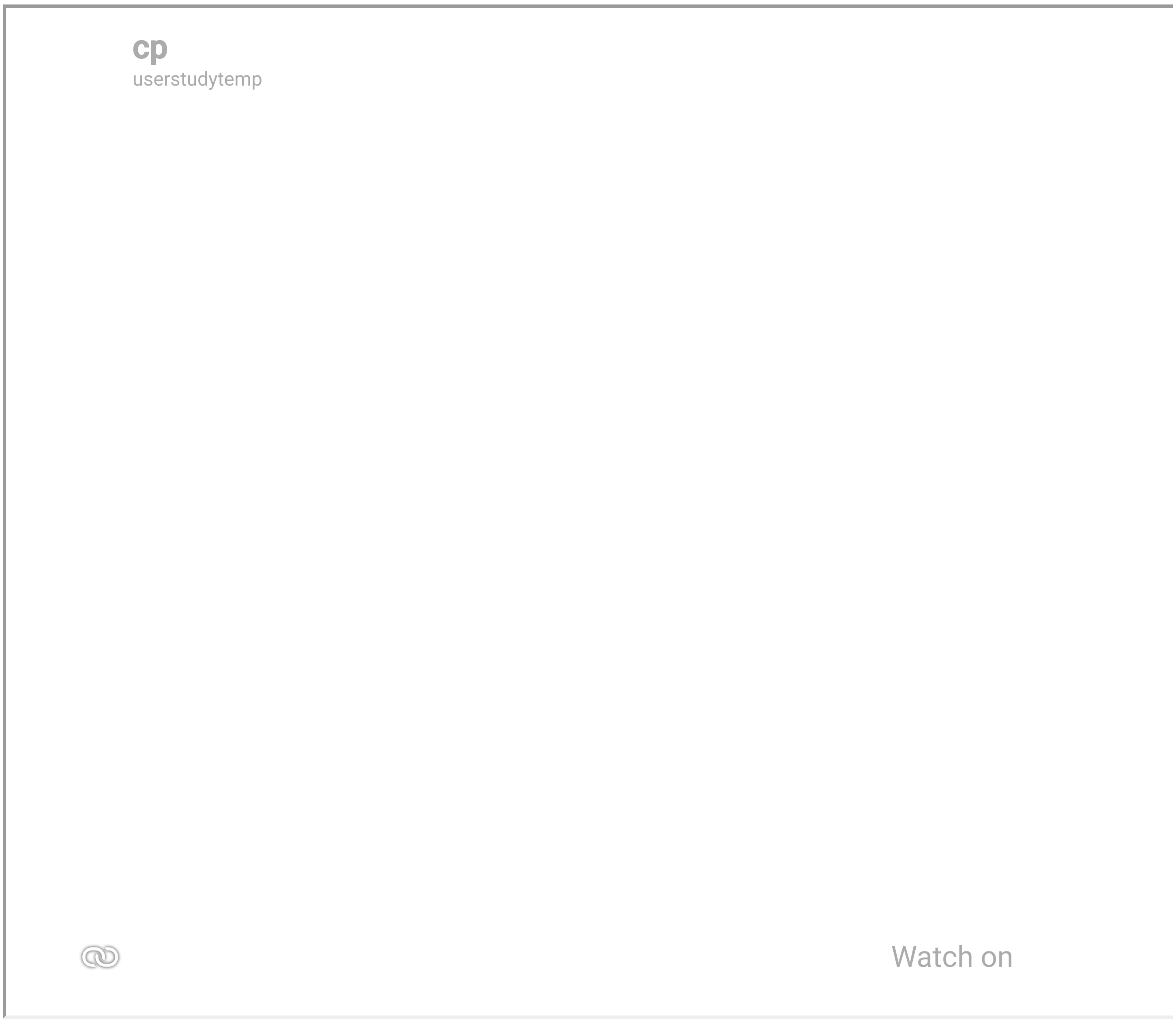
Transition to practice question

Now, please practice on the next question

practice

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.



Step 2:
Please answer the following questions carefully.

	Yes	No
Was there a straight road in the video?	○	○
Did the AV ever detect a straight road?	○	○
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect that it was approaching a stopped vehicle?	○	○
Was the AV driving straight partly because it detected a straight road?	○	○

	Yes	No
If there was a traffic jam ahead, would the AV have stopped?	○	○

Transition

Click Next To Begin The Real Survey

CP1

Step 1:
Please watch this video which shows the passenger's point of view in a self-driving car.

(note the car on the bottom left pointing out)

ccp1
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Is the other vehicle to the AV's immediate left affecting its willingness to move?	○	○

	Yes	No
When the light goes green, would the AV move faster if the car to its immediate left was gone?	○	○

CN1

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

ccn1
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Did the AV slow down partly because of the speed bump and stop sign?	○	○
If the speed bump and stop sign were gone, would the AV have moved faster?	○	○

AP1

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cap1
userstudytemp

Watch on

Step 2:
Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Did the AV mainly stop because it detected the pedestrian?	○	○

	Yes	No
If the pedestrian was not present at the end, would the AV still have stopped?	○	○

AN1

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

can1
userstudytemp

Watch on

Step 2:
Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Did the AV stop mainly because it detected it was approaching a stopped vehicle?	○	○

	Yes	No
If the stopped vehicle at the end was not in front of the AV, would the AV have moved further forward?	○	○

BP1

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cbp1
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever encounter a pedestrian?	○	○
Did the AV ever detect it was near a pedestrian?	○	○
Did the AV ever encounter a bike?	○	○
Did the AV ever detect it was near a bike?	○	○
Did the AV stop partly because it detected pedestrians?	○	○
Does the video make you significantly more wary of the AV than you were before watching?	○	○

BN1

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cbn1
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever encounter a pedestrian?	○	○
Did the AV ever detect it was near a pedestrian?	○	○
Did the AV ever encounter a bike?	○	○
Did the AV ever detect it was near a bike?	○	○
Did the AV stop partly because it detected the other cars in front of it and the red light?	○	○
If there were no other cars and the light was green, would the AV move forward?	○	○

AC1

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cac
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever encounter an intersection?	○	○
Did the AV ever detect an intersection?	○	○
Please select Yes to indicate you are paying attention	○	○
Please select No to indicate you are paying attention	○	○
Please select Yes to indicate you are paying attention	○	○
Please select No to indicate you are paying attention	○	○

CP2

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

ccp2
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Did the AV drive cautiously towards the end partly because it detected pedestrians nearby?	○	○

	Yes	No
If the red car was absent, would the AV have moved faster at the end?	○	○

CN2

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

ccn2
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Did the AV slow down partly because of the speed bump and stop sign?	○	○
If the speed bump and stop sign were not there, would the AV have moved faster?	○	○

AP2

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cap2
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○

	Yes	No
Was the pedestrian traversing the crosswalk at the end the main thing effecting the AV's speed?	○	○
If the pedestrian traversing the crosswalk at the end was not there, would the AV have behaved similarly?	○	○

AN2

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

can2
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever approach a stopped vehicle?	○	○
Did the AV ever detect it was approaching a stopped vehicle?	○	○
Was the AV ever very close to another vehicle?	○	○
Did the AV ever detect that it was very close to another vehicle?	○	○
Did the AV slow down mostly because it detected it was approaching a stopped vehicle?	○	○

	Yes	No
If there was no stopped vehicle in front of the AV at the end, would it have moved a bit further ahead?	○	○

BP2

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cbp2
userstudytemp

Watch on

Step 2:
Please answer the following questions carefully.

	Yes	No
Did the AV ever encounter a pedestrian?	○	○
Did the AV ever detect it was near a pedestrian?	○	○
Did the AV ever encounter a bike?	○	○
Did the AV ever detect it was near a bike?	○	○
Did the AV stop mostly because it detected pedestrians?	○	○
Does the video make you significantly more wary of the AV than you were before watching?	○	○

BN2

Step 1:

Please watch this video which shows the passenger's point of view in a self-driving car.

cbn2
userstudytemp

Watch on

Step 2:

Please answer the following questions carefully.

	Yes	No
Did the AV ever encounter a pedestrian?	○	○
Did the AV ever detect it was near a pedestrian?	○	○
Did the AV ever encounter a bike?	○	○
Did the AV ever detect it was near a bike?	○	○
Did the AV stop mostly because it detected the stopped vehicle in front of it?	○	○
If there was no stopped vehicle in front of the AV at the end, would it move a bit further ahead?	○	○

End

Thank you for your participation in the study, your responses will be used to help develop technology which improves people's situational awareness in self-driving cars.

For any questions, please contact ekenny@mit.edu

Click next to end the study.